Research Experiment Results

Hierarchical Federated Learning for Privacy-Aware, Low-Latency Multimodal IoT Systems (PP-HFM)

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*Complete record of every experiment executed. This revision adds the eICU-CRD external validation phase (§18–§20), the N-D clipping mechanism (§21) and the defect register (§22), and corrects three figures in the August edition. Sections marked **[NEW]** did not exist in that edition.*

1. System-Level Objective Evaluation

Objective	Target	Clinical	Wearable	Status
Uplink reduction	≥ 50%	90.82%	96.08%	Pass
Round latency improvement	> 0%	16.93%	21.68%	Pass
Composed privacy budget	≤ 1	0.970	0.966	Pass
Accuracy loss	≤ 2 pp	–0.92 pp	1.97 pp	Clinical pass; wearable inconclusive
P95 inference latency	< 100 ms	3.05 ms	1.22 ms	Pass
Successful updates	> 99%	100.00%	99.94%	Pass
Energy saving	≥ 20%	72.30%	76.84%	Pass

Interpretation. Seven of seven objectives pass on the clinical corpus, six of seven on the wearable with one inconclusive. The composed budget includes the conformal calibration release, which is stricter than reporting the training term alone. **Correction:** clinical P95 latency is 3.05 ms, read from `objectives_vitaldb.json`; the August edition printed 4.58 ms. The 16.36 ms in §16 is a different quantity — total pipeline P95 including preprocessing, not the objective's model-inference term.

2. Simulation Infrastructure

Simulator	Configuration	Measured result	Conclusion
NS-3 v3.46.1	9 IoT devices, 2 edge, 1 cloud; 54 Mbps access, 1 Gbps backhaul at 5 ms	Round latency 90.5 → 75.2 ms (clinical); 104.5 → 81.8 ms (wearable)	Edge tier reduces round latency by 16.93% / 21.68%
CloudSim Plus v8.5.6	IoT 300 MIPS / 1–3 W; edge 8000 MIPS / 50–200 W; GPU overhead on edge only	Centralised baseline 1266.25 J per round	Energy saving 72.30% / 76.84% against a 20% target

Interpretation. The energy figure quoted throughout is the CloudSim power-model result. A second figure of 93.5% exists from a linear bytes-to-Joules proxy, but under that model it is arithmetically identical to communication reduction and is therefore not an independent measurement.

3. Baseline Comparison

Baseline	Test accuracy	Test macro-AUC	Interpretation
B0 Centralized	0.5787	0.8211	Pooled-data reference
B1 Local-Only	0.4532	0.7511	Lower reference, no collaboration
B2 FedAvg	0.6543	0.8795	Best non-private baseline
B3 FedProx	0.6570	0.8747	No advantage over FedAvg here
B4 DP-FedAvg, ε ≈ 1	0.2159	0.6443	Collapses; the like-for-like DP reference
B5 MOON	0.3934	0.7418	Underperforms on this task

Interpretation. B4 is the critical row: conventional DP at the same budget destroys the model, which is what makes the head-only result meaningful. These baselines are not a fair comparator for the ε=1 head-only figure, because they train from scratch while the proposed runs start from pretrained frozen encoders. **§5 of this revision supersedes the causal reading**: the B4/B7/B6 decomposition shows the advantage comes from pretrained initialisation, not from restricting noise to the head.

4. Head-Only Differentially Private Fine-Tuning

Recipe	Noise scale σ	Test AUC	Round time
End-to-end DP-SGD	11.3281	0.562	1046 s
Head-only DP fine-tuning	11.3281	0.8742 ± 0.0072	50 s

Interpretation. The noise scale is identical in both rows, so noise was never the limiting quantity. What changed is the optimisation geometry: a near-linear probe over frozen pretrained features tolerates σ ≈ 11.3 while training from scratch does not. The privatised head is 560,645 of 2,157,766 parameters (26.0%). **Never write that freezing reduced the required noise** — σ depends on budget, steps and sampling rate, never on parameter count.

5. Privacy Budget Sweep, and What Actually Buys the Result

ε	σ	AUROC	AUPRC
0.5	5.92	0.8287	0.5110
1	3.24	0.8319	0.5464
2	1.88	0.8323	0.5381
4	1.21	0.8356	0.5590
8	0.85	0.8342	0.5544
∞ (no DP)	—	0.8158	0.5204

The B4/B7/B6 decomposition separates two explanations that the sweep alone conflates. B4 = full-model DP with random init; B7 = full-model DP with pretrained init; B6 = head-only DP with pretrained init.

ε	B4→B7 gap, clinical	B4→B7 gap, wearable
∞	+0.0140	+0.0023
8	+0.0549	+0.0879
4	+0.0735	+0.1310
2	+0.0943	+0.2079
1	+0.1105	+0.3088
0.5	+0.1274	+0.3948

Interpretation. Monotone across all five steps on both corpora. Privacy sensitivity, ε=∞ to ε=0.5: clinical B4 –0.1197 against B7 –0.0062 (**19×**); wearable B4 –0.3976 against B7 –0.0050 (**79×**). Wearable B4 falls to 0.5987, chance, while B7 holds 0.9936. **B7→B6 is approximately zero everywhere** (clinical –0.0053 to +0.0053). Head-only DP adds nothing to accuracy; its value is communication, since only head parameters are transmitted. Never quote the full-versus-head gap as evidence for head-only DP — it conflates initialisation with DP scope. **These points are single-seed; intervals are outstanding.**

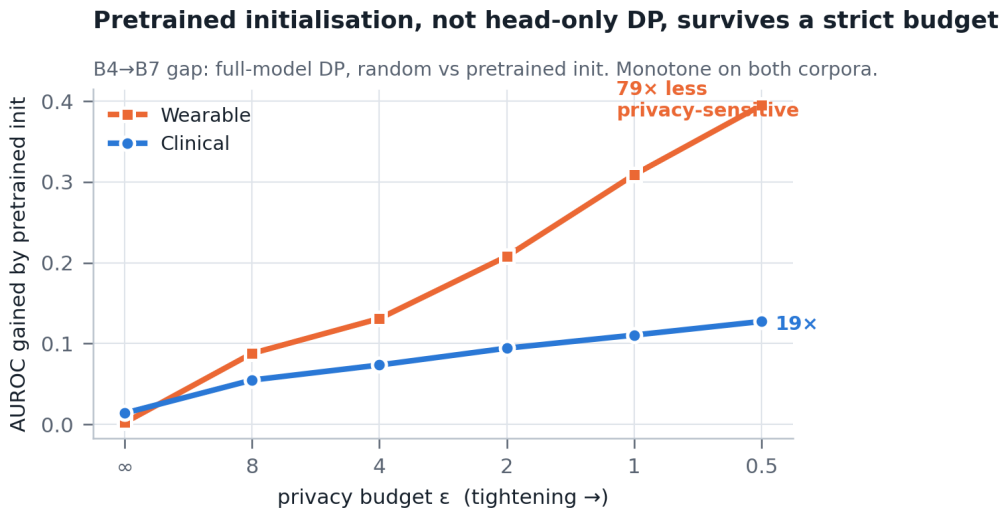


Figure 1. The advantage of pretrained initialisation widens monotonically as the budget tightens, on both corpora. If pretraining were merely a better starting point its advantage would be constant in ε; it is not. Single-seed points, so the claim is the monotone trend rather than any individual value.

6. Accuracy Cost at ε = 1

Dataset	Seeds	Accuracy cost	95% CI	Verdict
VitalDB (clinical)	5	–0.92 pp	[–2.82, +0.99]	Pass
WEAR (wearable)	15	1.967 pp	[1.760, 2.173]	Inconclusive

Interpretation. On the clinical corpus the point estimate is slightly negative, so the defensible claim is no measurable cost bounded by the interval, not zero cost. On the wearable corpus the interval straddles 2 pp, so the objective can be neither passed nor failed. The seed count was pre-registered at fifteen and deliberately not extended, since extending it after observing the result would constitute p-hacking.

7. Multimodal Fusion — Now Across Three Corpora [NEW]

Corpus	Modality A	Modality B	Best	Fused	Gain	Seed-sd	Verdict
VitalDB	0.8283	0.7680	0.8283	0.8653	+0.0370	5.9×	Fusion helps
WEAR	0.9976	0.9957	0.9976	0.9980	+0.0004	—	Ceiling; cannot test
eICU	0.6748	0.8212	0.8212	0.8249	+0.0037	0.2×	**Null**

Interpretation. All five seeds. eICU was the corpus nominated by the paper's own Discussion as the fair test — genuinely paired at source and difficult enough to leave headroom — and fusion is a null there, with AUPRC actually –0.0001. At one seed it read +0.0100 and the tooling printed *\u201cfusion HELPS\u201d*; five seeds dissolved it. This is the second time an n≤2 fusion figure proved to be seed luck, after the +0.0411/+0.1062 correction to +0.0370/+0.0671. **Only five-seed figures may be quoted.** The defensible claim is conditional: fusion pays on some tasks and not others, and the case must be made per task rather than assumed from the presence of two sensors.

8. Privacy-Unit / k-Scaling Experiment (VitalDB)

k (records/patient)	Record arm AUPRC	Patient arm AUPRC	Gap	p
1	0.5015	0.5067	–0.0052	0.75
2	0.5590	0.5084	+0.0507	0.0017
4	0.5924	0.5092	+0.0832	0.0015
8	0.5955	0.5089	+0.0866	0.0049

Interpretation. k=1 is a built-in null control: at one record per patient the record and the patient are the same unit and the arms must agree. They do. For k>1 the record arm rises while the patient arm stays flat (0.4967–0.5176 across every configuration), so the widening gap is the record arm improving, not the patient arm degrading. Extra records buy the patient arm nothing. AUROC follows the same shape (+0.0006, +0.0094, +0.0133, +0.0132) but no k reaches significance at three seeds. Accuracy is pinned at 0.8005 by an uncalibrated 0.5 threshold on imbalanced data and should not be quoted. **§21 tests whether the flat patient arm is an artefact of the clipping rule.**

9. Patient-Level versus Record-Level Differential Privacy

Quantity	Measured value	Interpretation
Corpus structure	48,712 windows across 6,257 cases; mean 7.79 per case	Every patient contributes more than one record
Patients contributing > 1 record	100%	The two guarantees differ for every individual
Cost of the unit upgrade	≈ 1.77 × σ	Direct accounting, against ε≈1→ε≈8 by conversion

Interpretation. Because not one patient contributes only a single record, a record-level and a patient-level guarantee are different statements for every individual present. This is the premise the project rests on, and it is measured rather than assumed. The 1.77× figure does not exist elsewhere in the literature for multimodal clinical FL.

10. Membership-Inference Audit

Arm	MIA AUC	Reading
Non-private control	0.6277	Positive control: attack signal present

Arm	MIA AUC	Reading
Record-level DP, ε≈1	0.5477	Small signal
Patient-level DP, ε≈1	0.5224	Near chance

Attack variant	Paired Δ (record – patient)	95% CI	p
Loss threshold	+0.0138	[–0.0034, +0.0310]	0.090
LiRA offline	+0.0325	[–0.0280, +0.0929]	0.210
LiRA online	+0.0034	[–0.0059, +0.0128]	0.365
LiRA online, global variance	–0.0211	[–0.0487, +0.0065]	0.101

Interpretation. Every interval includes zero. The predicted effect of approximately 0.20 lies far outside all four and is refuted. A small effect is not excluded — the half-width on the strongest attack is ±0.060 at five seeds. The correct wording is that the large predicted effect is refuted while a small one remains unresolved. On the patient arm LiRA adds nothing over a plain loss threshold (–0.0071), the signature of a defence with no difficulty-confounded signal left rather than of an underpowered adversary.

11. MIA Difficulty-Floor Control

Arm	Attack AUC	Difficulty floor	Margin
Positive control	0.6254	0.5626	+0.0627
Record DP, ε≈1	0.5477	0.5252	+0.0226
Patient DP, ε≈1	0.5224	0.5167	+0.0057
Record DP, ε≈16	0.5465	0.5743	–0.0278
Patient DP, ε≈16	0.5630	0.5259	+0.0371

Interpretation. Members come from released training patients and non-members from released test patients, and those pools differ in intrinsic difficulty (mean statistic 10.25 against 7.71), so an attack is otherwise rewarded for detecting difficulty rather than membership. The positive control still clears its floor, supporting attack validity; the ε≈1 arms clear theirs only slightly. This strengthens the refutation.

12. Membership Inference at a Loosened Budget (ε = 16)

Attack variant	Paired Δ	95% CI	Reading
Loss threshold	+0.0163	[+0.0123, +0.0204]	Significant, predicted direction, an eighth of predicted size
LiRA offline	–0.1788	[–0.2553, –0.1024]	Significant, reversed; over-correction artefact
LiRA online	+0.0057	[–0.0137, +0.0252]	Chance
LiRA online, global variance	–0.0552	[–0.0896, –0.0208]	Significant, reversed

Interpretation. At ε=16 the noise scale is thirteen times lower (σ 11.33 → 0.85), so a membership signal should survive if one exists. The hypothesis is refuted a second time under conditions maximally favourable to it.

13. Privacy Budget and Training Configuration

Configuration	Key result	Interpretation
Patient-level matched-budget DP	ε ≈ 0.97 composed	Meets the ε ≤ 1 target
Training + conformal calibration	0.7705 + 0.2000 = 0.9705	Composition remains below 1
Patient-level upgrade	≈ 1.77 × σ	Measurable optimisation cost
Head-only DP	AUC 0.8742	Parameter-efficient private adaptation is effective
Rigorous public pretraining	–0.022 AUROC against oracle	The cost of an end-to-end-valid guarantee

Operating point, and a drift that was caught: `--epsilon 0.80 --epsilon_conf 0.20 --batch_size 64` *composes to 0.9705 (clinical) and 0.9656 (wearable). The driver had drifted to 1.0 / 0.5 / 16, composing to 1.457 — a failing budget — and was corrected on 8 September 2026.* `epsilon_spent` *in a results CSV is the TRAINING term only; the reportable number is* `epsilon_TOTAL` *in* `privacy_total_*.json`.

14. Ablation Experiments

Removed component	AUROC	Uplink	Conclusion
None (full system)	0.8541 ± 0.0035	251.97 MB	Reference configuration
Hierarchy	0.8484 ± 0.0057	50.40 MB	Supports systems efficiency, not accuracy
Privacy	0.8307 ± 0.0060	251.97 MB	Removing DP makes the model worse
Sparsification	0.8553 ± 0.0015	1259.75 MB	+0.0012 AUROC for 5× the traffic
Quantisation	0.8543 ± 0.0036	403.14 MB	+0.0002 AUROC for 1.6× the traffic
Multimodality	0.8283	—	See §7
Added ASWA aggregation	0.8542 ± 0.0037	251.97 MB	+0.0005 AUROC; no contribution

Interpretation. Removing DP makes the model worse, confirming §5 — DP regularises rather than taxes on this task. Both compression stages earn their place. ASWA contributes nothing and was deleted. **Caveats:** only the privacy arm (C2) resolves at three seeds and its sign flips between corpora, so component necessity is not established; and the ASWA decision rests on a single seed, which is a judgement rather than a measurement. Separately, the seed-paired re-analysis puts the hierarchy effect at –0.005 AUROC (clinical p=.032, wearable p=.0018), not the +0.096 an unpaired reading suggested.

15. Communication and Compression

Measurement	Bytes per round	Interpretation
Full-model FP32 FedAvg	258,931,920	Reference
Head-only, uncompressed	67,277,400	Effect of freezing the encoders
Observed, after defect fix	16,821,300	90.82% reduction against the reference

Interpretation. A defect was found and corrected: the compressor took a fixed fraction of every tensor, so frozen encoders — whose delta is exactly zero — still contributed 20% of their elements as explicit zeros. Omitting all-zero tensors accounted for 78.87% on its own. The fix provably did not alter learning: the dense path is bit-identical over five rounds and an isolated re-run moved AUC by 0.0001.

16. Inference Latency and Edge Deployment

Precision	Provider	Model size	Total P95	Status
FP32	CPU	10.64 MB	16.363 ms	Pass; recommended
FP32	GPU (CUDA)	10.64 MB	22.476 ms	Pass
INT8	CPU	6.55 MB	31.028 ms	Pass but slower than FP32
INT8	GPU (CUDA)	6.55 MB	73.150 ms	Pass but substantially slower

Interpretation. The privacy apparatus is free at serving time: conformal set construction occupies 0.68% of the inference path and the private mechanism occupies none of it, since its noise is applied to gradients during training. INT8 is a characterised negative — 1.90× slower on CPU, 3.25× on GPU, and it misses the 4 MB target at 6.55 MB. Dynamic quantisation accelerates only matrix multiplies while this model is dominated by convolutions and a recurrent path; on CUDA the quantised ops have no kernel and fall back to CPU, with 3180 memory-copy nodes inserted. Deploy the FP32 artifact.

17. Feature Attribution

Recipe	Share of attribution carried by missingness masks
Full-model DP	28.2%
Patient-level DP	22.0%
Record-level DP	21.6%
Full-model, no DP	19.9%
Head-only, no DP	15.8%

Interpretation. Differential privacy shifts attention toward missingness, monotonically in how much privacy is applied: under noise the model relies more on which measurements exist than on their values. Rank correlation is 0.95 between head-only private

and non-private recipes. This is **associational**: it does not establish that accuracy would fall without the mask channels, which only a mask ablation could show.

18. eICU-CRD External Validation — Corpus and Gates [NEW]

The standing objection to this work was that every privacy claim rested on one dataset from one institution whose clients were a partition we invented. eICU-CRD answers it with clients that are real hospitals taken as given.

Property	Value	Why it matters
Windows / individuals / stays	3,782,805 / 132,736 / 185,140	Scale sufficient for patient-level claims
Clients	187 (206 hospitals; 6 pooled for size, 14 for zero positives)	**Corrects the 208 quoted in planning documents**
Individuals per client	median 444	WEAR has 1, where the units coincide and the question cannot arise
Records per individual	28.50 mean	Extends k beyond VitalDB's reach
Mortality	0.0422 (pre-registered forecast 0.0260)	Disclosed as a forecast miss, not silently updated
Encoder pretraining	RIGOROUS — 526,867 indices verified disjoint from every client	**eICU utilities carry no oracle caveat, unlike every VitalDB figure**

Gate	Result	Verdict
Corpus validation	Pairing, identity, split overlap 0, individual-disjoint; records/individual 28.50; individuals/client 444	PASS, no CAPACITY failure
Modality kill switch	Best single modality 0.8212 against a ~0.95 abandon threshold	PASS — corpus not disqualified

19. eICU k-Scaling — the Pre-Registered Test [NEW]

A numeric forecast was fitted on VitalDB alone and frozen, dated, before the corpus was built: $\text{gap}(k) = A(1 - e^{-b(k-1)})$, A = 0.0878 ± 0.0035, b = 0.883 ± 0.129, RMSE 0.0028, acceptance band ±0.030.

k	Record AUPRC	Patient AUPRC	Observed gap	95% CI	p	Predicted	In band
1	0.2105	0.2212	–0.0107	[–0.0317, +0.0103]	0.159	0.0000	**PASS** (control)
2	0.2245	0.2185	+0.0060	[–0.0041, +0.0161]	0.122	0.0515	MISS
4	0.2318	0.2208	+0.0110	[–0.0603, +0.0824]	0.575	0.0816	MISS
8	0.2374	0.2239	+0.0135	[–0.0455, +0.0725]	0.429	0.0876	MISS
16	0.2363	0.2258	+0.0105	[–0.0249, +0.0459]	0.331	0.0878	MISS

Interpretation. P1a passes. The control is –0.0107, inside the ±0.020 tolerance fixed in advance, so the harness does not separate the arms where the construction says it cannot and the remaining rows are admissible. This was a registered precondition: had it failed, no eICU result would be reported at all. **P1b is refuted.** Three of four tested k fall outside the band and the measured cost is roughly a sixth of predicted. **The shape survives** — negative at k=1, positive from k=2, rising monotonically. The registered reading of exactly this outcome, fixed in advance, is that the shape is structural and the magnitude is corpus-dependent. **Power:** nothing resolves at three seeds; every interval covers zero, no p below 0.12, and resolving the observed k=8 effect would need about fifty seeds against the three fixed in advance. Extending them now would convert a registered test into a search, so it is not done. AUROC gaps are –0.0012, –0.0015, +0.0018, +0.0001 — no trend at all, where VitalDB showed a rising one.

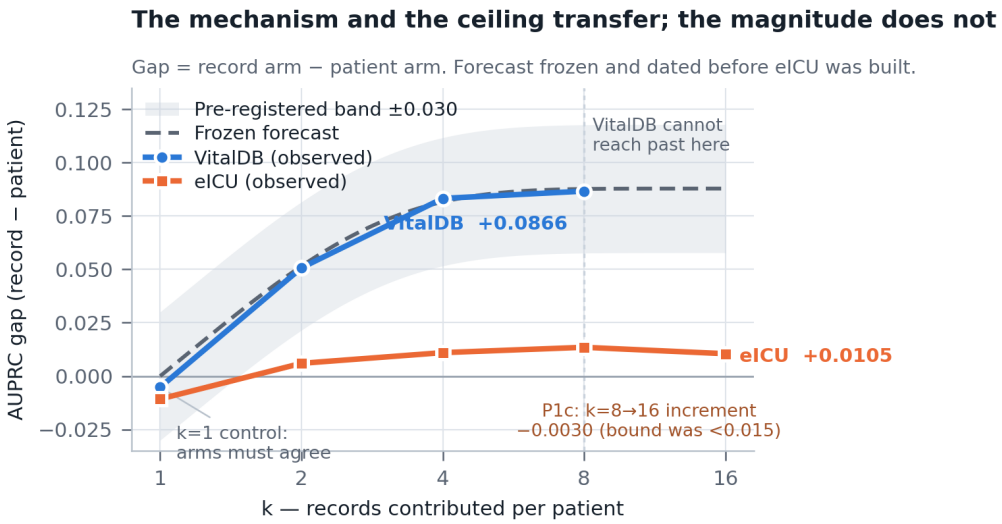


Figure 2. Both corpora reproduce the ordering the mechanism predicts — negative at k=1 where the units coincide, positive and rising thereafter — but eICU sits far below the band frozen before the corpus was built. The k=1 control is what licenses reading the rest of the curve as an effect of the privacy unit rather than of the apparatus.

Registered prediction	What it required	Measured	Verdict
P1a control	|gap| < 0.020 at k=1	–0.0107	**PASS**
P1b point forecasts	within ±0.030 of the frozen curve	~1/6 of predicted	**REFUTED**
P1c saturation	gap beyond k=8 exceeds gap at k=8 by < 0.015	increment **–0.0030**, CI [–0.0316, +0.0255], p=0.692	**SATISFIED**
P1e asymmetry	individual-arm AUPRC range < 0.020 across all k	range **0.0073** (0.2185–0.2258) while the record arm rises 0.2105→0.2374	**SATISFIED**
P1f AUROC	same sign, underpowered	–0.0012, –0.0015, +0.0018, +0.0001, +0.0105	No trend over k≤8

P1c is the prediction VitalDB structurally cannot make, since it cannot reach k > 8, and it is the stated reason this corpus was built. It holds: the gap stops climbing, with the point estimate on the favourable side of zero. Together with P1e this means **what differs between the corpora is the size of the effect, not its mechanism** — the registered reading of exactly this outcome, fixed in advance.

A confound on P1b, found 2026-09-21 and disclosed as amendment A2. `k_scaling.py` never passed `--epsilon_conf`, so every k-scaling run inherited the trainer's default of 0.5 rather than the registered operating point of 0.20. Two consequences, of unequal seriousness.

Grid	ε_{train}	ε_{conf}	ε_{TOTAL}
VitalDB k-law — the forecast was FITTED here	0.9935	0.5	1.4935
eICU k-law — the forecast was TESTED here	0.7956	0.5	1.2956

The reported budget is wrong in both: **no k-scaling result may be described as measured at ε ≤ 1.** The conformal term is spent after training on the calibration release and never enters the DP-SGD noise, so it corrupts what was reported rather than what was measured. Fixed at source; the flag now defaults to 0.20.

The comparability defect is the material one. The two corpora trained at different budgets: the forecast was fitted at $\varepsilon_{train} = 0.9935$ and tested at 0.7956, because the eICU driver used the registered operating point while the original VitalDB grid used this script's default of 1.0. eICU therefore trained under MORE noise, and more noise compresses the difference between arms — the same direction as the observed shortfall. **P1b's refutation is confounded and is not reported as a clean corpus effect.** P1a, P1c and P1e are unaffected: each is internal to eICU and compares arms trained at one identical budget. The VitalDB grid is being re-run at the registered operating point so the comparison is like-for-like; the forecast is not refitted and both grids will be reported.

k = 32 is reported untested. *Reaching 32 records requires multiple admissions, so only 20.2% of individuals qualify and they are systematically sicker; k=16 retains 91.0%. Recorded as dated amendment A1 to the registration.*

20. Why Fusion Pays on One Corpus and Not Another [NEW]

Corpus	Headroom (1 – best)	Imbalance |A–B|	Gain	% of headroom captured
VitalDB	0.1717	0.0603	+0.0370	21.5%
WEAR	0.0024	0.0019	+0.0004	15.0%
eICU	**0.1788**	0.1463	+0.0037	**2.1%**

Interpretation. The headroom explanation is refuted. eICU has *more* headroom than VitalDB and captures a tenth as much of it. Ceiling effects explain WEAR and fail on eICU, so they cannot be the general account. This explanation was proposed during analysis and is recorded as withdrawn rather than dropped.

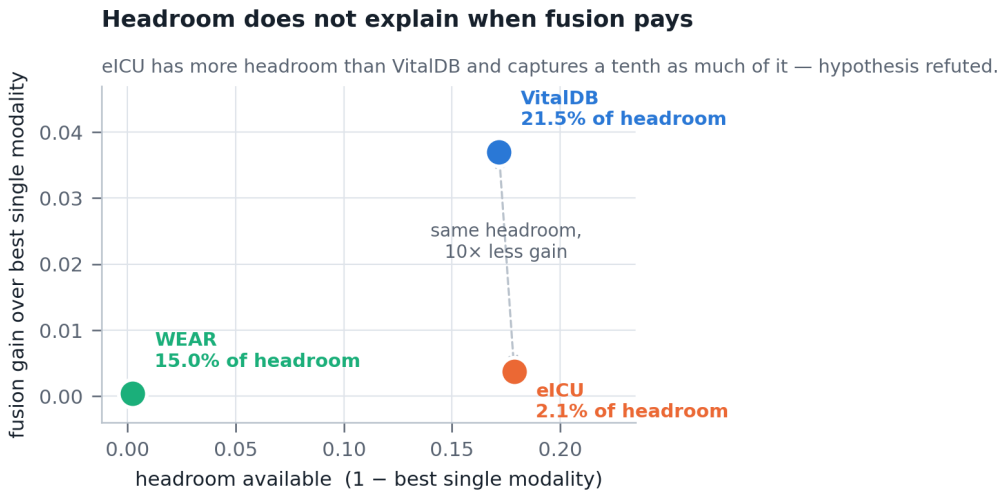


Figure 3. *If headroom governed fusion, the three corpora would lie on a rising line. eICU breaks it: essentially the same headroom as VitalDB, a tenth of the gain. WEAR is consistent with the hypothesis and eICU is not, so headroom cannot be the general account.*

The surviving hypothesis is modality balance — fusion pays when the second stream is competitive, not merely present. Three observational corpora cannot separate that from any other property of eICU, so it was tested by manipulation within eICU: zeroing the APACHE severity score from modality B, holding corpus, partitions, encoders, seeds and protocol fixed. Predictions F1–F4 were frozen first.

What the corpora actually pair matters here, and it is easy to overstate the eICU result. Modality B is a 2048-d two-stream video feature on WEAR and a 34-d clinical and laboratory vector on VitalDB, but on eICU it is **five admission scalars** — age, sex, weight, height and APACHE. That is metadata, not a second sensing stream, so the eICU null tests whether 24~h of vitals add to a severity score rather than whether multimodal sensing helps.

Arm	AUROC	Prediction	Outcome
a (vitals)	0.6973 ± 0.0048	—	—
b (tabular, APACHE zeroed)	0.5947 ± 0.0031	F1: below 0.75	**HELD** (from 0.8212)
Imbalance	0.1026	F2: below 0.08	FAILED
both (fused)	0.6911 ± 0.0212	F3: gain ≥ +0.015	**FAILED** (gain –0.0062)

And APACHE is a clinician-computed score built to predict in-hospital mortality — the label itself. Zeroing it collapses modality B from 0.8212 to 0.5947, so that single field carries almost all of B's signal. The eICU fusion question therefore reduces to whether vitals add anything to a validated mortality predictor, and the answer being no is a statement about the benchmark rather than about multimodality. This is the transferable finding: **fusion cannot pay when one modality already contains a purpose-built predictor of the target**, and a multimodal benchmark should be checked for that before its fusion result is believed.

> **Interpretation.** F1 held emphatically — one severity score carried almost all of modality B. But the lever **overshot**: it pushed B below A rather than level with it, so balance was never achieved (0.146 → 0.103, against a 0.08 target) and F2 failed. Under the registered outcome table that is *\u201clever wrong, inconclusive\u201d* rather than *\u201cbalance refuted\u201d*. A matched control on the current partitions is running, required because a 20% training-set change (the disjoint pretraining carve-out) would otherwise confound the comparison — recorded as amendment F-A1.

21. N-D — Normalised Patient Clipping [NEW]

In `model/privacy/patient_dp.py` a patient's gradient is summed over that patient's k windows and then clipped to a fixed norm C. The pre-clip norm therefore grows with k, so patients with long recordings saturate the clip and are systematically down-weighted. **This is the suspected mechanism behind the flat patient arm in §8.** If so, the central cost figure is partly an artefact of a clipping choice rather than an inherent property of the guarantee.

k	Pre-clip norm, none	Clip binds?	Norm under /k	Binds?	Norm under /√k	Binds?
1	0.250	no	0.250	no	0.250	no
4	1.000	no	0.250	no	0.500	no
8	2.000	**yes**	0.250	no	0.707	no
16	4.000	**yes**	0.250	no	1.000	no
32	8.000	**yes**	0.250	no	1.414	**yes**

Interpretation. The guarantee is unchanged, and this must be stated precisely or the mechanism reads as a privacy cheat. Sensitivity is set by the clip, not by what is clipped: after rescaling, every patient's contribution is still bounded by C, so removing one patient still moves the sum by at most C. σ, ε and δ are identical across arms — verified by inspection, since `clip_norm` appears in no noise-multiplier or accountant code path. The arms compare at a matched budget and the difference is optimisation geometry alone. This is the same argument §4 makes for head-only DP.

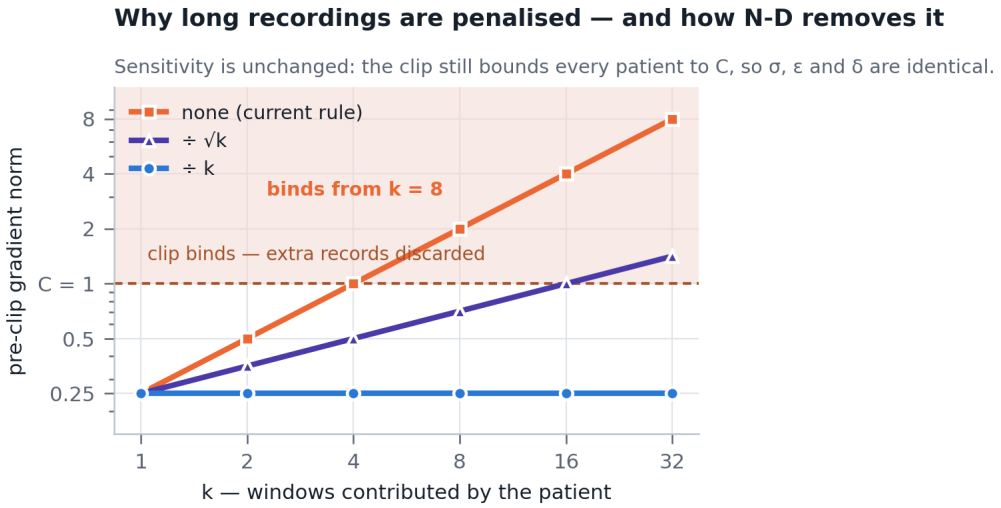


Figure 4. *Under the current rule a patient's pre-clip gradient norm grows with k, so from k=8 the clip binds and the additional records are discarded — the suspected cause of the flat patient arm in §8. Dividing by k removes the dependence entirely; dividing by √k defers it to k=32. Illustrative scale; the measured saturation is reported by the new diagnostic.*

Split	windows/patient sd	none	k	sqrt_k	N1
VitalDB, k=8 partition	0.67	0.6847	0.4069	0.5556	FAILED
eICU, natural partition	18.13	0.3945	0.2039	0.2939	FAILED

N1 failed on both, and is recorded as failed rather than rewritten. Two findings explain it and both are kept. First, **VitalDB cannot test N-D at all**: its natural partition is identical to the k=8 capped one — min 3, median 8, max 8, sd 0.67 — so nearly every patient contributes exactly eight windows and there is no k-dependence to correct. The VitalDB N-D utility numbers are therefore not reported as a test of N-D. eICU is ~21x more heterogeneous (2 to 192 windows per patient).

Second, on eICU, **saturation rises with k but k is not its dominant driver**. Across $2{,}923$ per-patient gradients at $C = 1$:

k bin	patients	mean pre-clip norm	saturation
1–4	27	1.784	0.148
5–12	125	2.708	0.216
13–24	1,937	61.838	0.384
25–48	605	58.622	0.436
49+	229	69.070	0.502

The direction N-D predicts is confirmed and highly significant — Spearman $\rho(k, \text{norm}) = +0.245$ at $p = 2.5\times10^{-41}$, and patients with the most records are truncated at 0.502 against 0.148 for those with fewest, a $3.4\times$ difference. But $\rho(k, \text{saturated}) = +0.138$ and mean norms of $60+$ against $C = 1$ show most patients saturate for reasons independent of k, which is why dividing by k halves saturation rather than removing it. **The clipping penalty is real, measured and significant; it is secondary rather than dominant.** That stands as a mechanism finding for the flat individual arm in §8 without any utility run.

Implemented 20 September 2026 as `--clip_norm {none,k,sqrt_k}`, default `none` so every published result is bit-identical, with a new `clip_saturation` diagnostic. Predictions frozen before running: N1 saturation >0.50 under *none* and <0.10 under *k*; N2 patient-arm AUPRC rises ≥0.02; N3 the k=8 gap shrinks ≥0.02. **If N1 holds and N2 fails, the cost is inherent to the guarantee, which strengthens the existing claim** — that outcome is as reportable as success. Experiment queued; VitalDB k=8, three seeds, three modes.

22. Defects Found and Fixed [NEW]

Defect	Effect if unfixed	Status
Training collapse on long epochs. Model peaks near step 1,000 then decays to constant output (val AUROC exactly 0.5000, prediction sd 0.00000 by step 5,000). Validation ran once per epoch; an eICU epoch is 41,394 steps.	Every eICU number silently wrong. Worst case: it also affected encoder pretraining, whose output k-scaling *freezes*, so collapsed encoders would have propagated into every k-scaling result without raising an error.	Fixed in 3 trainers via opt-in `--eval_every / --val_cap / --max_steps`; defaults reproduce the original cadence exactly, so published results are unchanged
eICU individuals span two labels. A uniquepid can survive one admission and die in a later one — 1.46% of train, 1.38% val, 1.41% test individuals. Stays have zero conflicts.	Assertion failure blocking all 36 k-scaling runs; person-level metric aggregation undefined.	Registry field `metric_group_file`; eICU groups metrics by stay while the DP unit and split key remain the individual
Validity message conflated *no paired interval* with *not executed*	The k=1 validity anchor reported itself as not run	Patched
k_scaling never passed `--epsilon_conf`. Every k-scaling run inherited the trainer default of 0.5 instead of the registered 0.20.	Reported budgets of 1.4935 (VitalDB) and 1.2956 (eICU), both above the ε ≤ 1 objective; and the forecast was fitted and tested at different training budgets.	Flag now passed, default 0.20; VitalDB grid re-running at the registered point (amendment A2)
`clip_saturation` was returned by the DP routine but consumed by nobody	The N-D mechanism gate was unmeasurable from the pipeline	Wired through to the per-round CSV
Nine argparse gates rejected `eicu`	eICU unrunnable	Resolved from REGISTRY

23. Negative and Corrected Results

Item	Outcome
Predicted patient-level MIA vulnerability	Refuted at ε=1 and again at ε=16, with a passing positive control both times
Adaptive per-group privacy budget (SADP)	Underperforms uniform DP at equal ε by construction; cross-group composition drives σ from 11.3 to 70
INT8 edge quantisation	Slower than FP32 on both providers; misses the size target
ASWA aggregation	+0.0005 AUROC; deleted from the codebase (single-seed decision)
Fusion gain at two seeds	Corrected downward at five seeds; AUPRC fell 37%
Fusion on eICU	**Null at five seeds** on the corpus nominated as the fair test
Headroom explanation for the fusion null	**Proposed, tested, refuted, withdrawn**
P1b point forecasts (eICU)	**Refuted — 3 of 4 outside the band; magnitude ~1/6 of predicted**
P4a hospital heterogeneity	**Refuted — real hospitals span 0.0–18.3% against VitalDB's 3.2–50.7%**
Legacy corpus	Withdrawn entirely; modalities paired by a construction rule rather than at source
Wearable accuracy objective	Inconclusive; seed count pre-registered and deliberately not extended
Wearable fusion	At ceiling; structurally unable to corroborate the multimodal claim
Head-only DP fine-tuning	Memorises members harder than full-model training, 0.6277 against 0.5122 — a caveat to our own recipe

24. Experiments Not Yet Run [NEW]

Experiment	Why it matters	Status
k = 16 saturation (eICU)	Tested P1c, the one prediction VitalDB cannot make	**DONE** — P1c SATISFIED
Fusion matched control	Guarded the balance test against a 20% partition change	**DONE** — null is robust
Fusion balance, titrated lever	Stage 1 overshot; a scaled lever can hit a balance target	**Queued**, ~2 h
N-D clip normalisation	The only candidate for an original mechanism rather than a measurement	**Running**, 7/9
Conformal coverage validation	ε_{conf} = 0.20 is a fifth of the reported budget and its coverage has never been measured	Not run — ~3 h
Multi-seed B4/B6/B7	The 19×/79× headline is single-seed, with no intervals	Not run — ~1 day
Rigorous pretraining, VitalDB/WEAR	Would remove the oracle caveat from those corpora as it has from eICU	Not run — optional
Mask-channel ablation	Would turn §17's associational evidence into proof	Not run — optional
S3 hierarchical, no DP	Completes the topology axis	Not run — optional
k = 32 (eICU)	**Reported untested, not dropped** — only 20.2% of individuals reach 32 records and they are re-admitted patients	Will not be run; amendment A1

25. Experiment Coverage Across Corpora [NEW]

The same experiment was not run on every corpus, and the paper must not imply that it was. Read from the artefacts on disk rather than from intent. Each corpus was built to exercise a different property, and exactly one experiment — the multimodal fusion ablation — ran on all three under an identical protocol.

Experiment	Script	VitalDB	WEAR	eICU
Objectives O1–O4	check_objectives.py	yes	yes	**no**
Baselines B0–B5	baselines.py	yes	yes	**no**
ε sweep B4/B7/B6	epsilon_sweep.py	yes	yes	**no**
O2b multi-seed	epsilon_sweep.py	yes	yes	**no**
k-scaling	k_scaling.py	yes	impossible	yes
Fusion ablation	modality_ablation.py	**yes**	**yes**	**yes**
MIA / LiRA	lira_attack.py	yes	**no**	**no**
Component ablation	component_ablation.py	yes	yes	**no**
Latency / ONNX	profile_model.py	yes	yes	**no**

Every script above resolves `--dataset` from the registry and accepts eICU. **The gaps are compute, not capability** — with one exception: WEAR cannot run k-scaling for a structural reason, not an omission. Each of its clients is a single participant, so client-, user- and individual-level guarantees coincide and the privacy unit cannot vary.

25.1 What each gap would cost, measured

Costs below are extrapolated from measured wall-clock: a ten-round run over eICU's full 2,122,350-window training split takes approximately 4.5 h on the available GPU.

Gap	Why it is missing	Cost to fill
eICU O3a latency, O3b reliability, O1b, O4 energy	No training needed — ONNX export and the network and energy simulators run from an existing checkpoint, and eICU checkpoints exist from the k-scaling grid.	**minutes to ~1 h**
eICU O1 uplink, O2a budget	One *proposed* run at the registered operating point over the full corpus, which the cheap stages above then read.	~4.5 h
eICU O2b accuracy cost	Paired ε=1 against ε=∞, five seeds. Arrives with fewer seeds and rounds than WEAR's fifteen at thirty, so the comparison would be uneven.	~45 h
eICU component ablation	Seven arms over the full corpus.	~31 h

Gap	Why it is missing	Cost to fill
eICU ε sweep B4/B7/B6	Six budgets x three recipes x seeds.	~81 h per seed
eICU MIA / LiRA	Sixteen shadow models, each a full training run.	~72 h
WEAR MIA	Possible but near-degenerate: one individual per client, so a patient-level attack has no unit to separate.	not informative
WEAR k-scaling	**Structurally impossible**, as above.	cannot be filled

25.2 What this means for what may be claimed

Three sentences follow from the table and should appear in the manuscript rather than being left for a reviewer to derive. **First**, the system-level objectives are demonstrated on two corpora, not three, and eICU — the only corpus whose encoders are disjoint and whose utilities are therefore valid end to end — carries no objectives table. **Second**, the privacy audit is single-corpus: every membership-inference result comes from VitalDB, and the pre-registered prediction P5a that the head-only memorisation caveat would reproduce on eICU was never tested. **Third**, the multimodal comparison is the one claim with complete like-for-like coverage, which is precisely why it is stated as a per-task criterion rather than a universal benefit: the coverage is complete, and it says the answer depends on the pairing.

26. Summary of Outcomes

Class	Count	Items
Objectives met	13 of 14	Seven clinical, six wearable; one inconclusive by design
Positive findings	7	1.77× unit cost · pretrained init 19×/79× · k-law with built-in control · compression free · hierarchy resolved · DP regularises · privacy apparatus free at serving time
Refuted predictions	4	Patient-level MIA advantage (twice) · P1b magnitude · P4a heterogeneity · headroom explanation of the fusion null
Characterised negatives	5	SADP · INT8 · ASWA · fusion on eICU · head-only DP memorisation caveat
Defects found and fixed	8	4 ablation defects (Aug) · compressor zero-padding · training collapse · label conflict · validity message
Pre-registrations	3	eICU k-law · fusion balance · N-D; two amendments recorded

The pattern that matters. *Four predictions were registered in advance and refuted, each with a passing positive control alongside it, and one explanation was proposed during analysis and withdrawn in writing when it failed a test. A paper whose negative results are as carefully instrumented as its positive ones is the argument for trusting the positive ones — and on this record the central measurement, the cost of a patient-level guarantee, is supported by a self-validating design on two independent clinical corpora.*